Untying Prefill and Decode Loops in Looped Language Models

Yang Hu

Abstract

Looped language models reuse one block of layers several times per token and, under fixed-depth inference, run the same number of loops when they prefill the prompt and when they decode each answer token. We untie the two. A small *loop adapter* trained on a frozen looped model lets it prefill with m loops and decode with n: one set of parameters serves all 16 pairs up to four loops, and one stored prompt KV cache serves every decode loop. On the checkpoints and benchmarks we test, the two counts buy different things: *prefill loops mainly raise knowledge and long-context benchmarks (MMLU, MMLU-Pro, RULER, HELMET), decode loops mainly raise multi-step math (GSM8K, MATH-500)*, with exceptions that depend on the task and its prompt. On Ouro-2.6B-Thinking, prefill 4 / decode 2 scores at least as high as the native two-loop model on all eight standard benchmarks we test (significantly higher on five), at the same decode cost, and within two points of the native four-loop model on six; on GSM8K it is 1.0 point below the four-loop model on the full test set and, on 64 timed problems, answers in $0.55\times$ its time. Untying the loops also makes looped decision models more accurate when they decide with one loop, against a control continued with the same data and updates. Code, results and per-item outputs: `https://github.com/yanghu819/untying-prefill-decode`.

1 Introduction

Looped language models such as Ouro (Zhu et al., 2025), Huginn (Geiping et al., 2025) and Mixture-of-Recursions (Bae et al., 2025) obtain depth without adding parameters by running a shared block of layers several times per token (Dehghani et al., 2018). Under the fixed-depth inference we study, the price is paid twice: every generated token runs every loop, and every loop keeps its own key–value (KV) cache.

Yet the two phases of inference are not alike. Prefill is parallel and paid once per request; decoding is sequential and paid for every token. A looped model nevertheless runs the same number of loops in both. What do prefill loops buy, and what do decode loops buy? If the answers differ, the two counts should be chosen separately.

We write prefill m / decode n for prefilling the prompt with m loops and decoding every answer token with n loops, and native n/n for the unmodified model, which uses n loops for both. When $m \neq n$, the decode loops must attend to prompt KV produced by a different loop than their own. Our *loop adapter* combines LoRA on the frozen model with a learned map on the prompt KV for every decode loop (Figure 2); one stored prompt cache serves every decode loop. We train one loop adapter and evaluate it at all 16 pairs (m, n) with its parameters fixed.

The resulting grid shows a division of labour (Figure 1). On the benchmarks we test, prefill loops mainly move knowledge and long-context scores (MMLU, RULER-16K), and decode loops mainly move multi-step math (GSM8K). With the decode loops held fixed, more prefill loops help on five checkpoints from two model families (Table 1).

The best pair therefore depends on the task, and an untied model can spend its loops where they pay (Figure 4): on Ouro-2.6B-Thinking, prefill 4 / decode 2 scores at least as high as native 2/2 on all eight benchmarks at the same decode cost (significantly higher on five), and within two points of native 4/4 on six. On full GSM8K it scores 93.3 against 94.2 for native 4/4; on 64 timed problems its answer time, prefill included, is $0.55\times$ that of native 4/4 (Figure 7). Against measured decode throughput, the loop adapter is more accurate than native models at one and two decode loops and faster at three and four (Figure 5). The same untying carries over to looped decision models: against

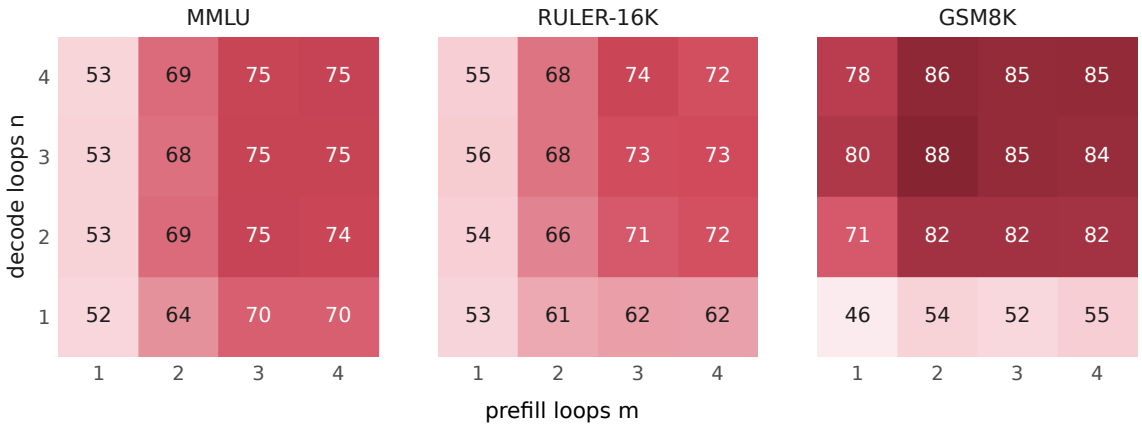


Figure 1: **Prefill loops raise knowledge and long-context accuracy; decode loops raise multi-step math.** One loop adapter on frozen Ouro-2.6B evaluated at all 16 (prefill m, decode n) pairs. Knowledge (MMLU) and long-context retrieval (RULER-16K) improve with m; multi-step math (GSM8K) improves with n. Grid subsets: MMLU 20 questions per subject, RULER-16K 260 rows, GSM8K 400 problems; cells show accuracy (%), one colour scale for all panels; full-set comparisons are in Table 2.

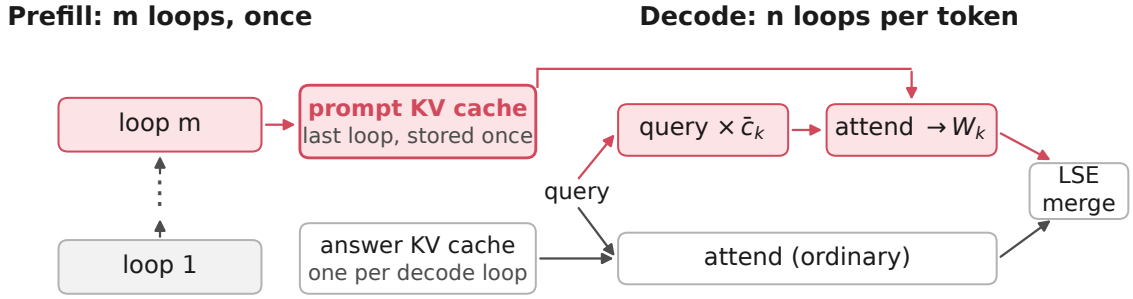


Figure 2: **The loop adapter.** The frozen model prefills the prompt with m loops and keeps one prompt KV cache. Each of the n decode loops reads that cache through its own map: a learned scale on each rotary (RoPE) pair of the keys, which can be moved onto the query so the cache is never rewritten, and a value map applied after attention. Attention over the prompt and over the generated tokens is computed separately and combined into one softmax, as in blockwise (FlashAttention-style) attention. The prompt KV is stored once for every n: total KV $b(\ell + n\,t)$ instead of $b\,n(\ell + t)$, with b bytes per token and loop and ℓ, t the prompt and answer length.

a control continued with the same data and updates, Lev models that prefill their context with more loops are 4.4–7.5 points more accurate at one decision loop on three backbones (Figure 6).

Our contributions are a measurement of what prefill loops and decode loops each buy in looped language models, a loop adapter that unties them on existing checkpoints with an exact single prompt cache, and an end-to-end measurement of the time it saves. Code, per-item results and a script that recomputes the paper's statistics accompany it.

2 Related Work

Looped and elastic depth. Accordion (Zhang et al., 2026) and Elastic Latent Recurrence (Ladhak et al., 2026) train one model for several depth budgets; ELR also finds that keeping only the last loop's KV retains most of the gain. The Parallel Loop Transformer (Wu et al., 2025a), LoopCoder-v2 (Yang et al., 2026) and shared-memory looped transformers (Monea et al., 2026) let later loops read the first loop's KV plus a short local window. All run the same number of loops for prefill and decode.

Phase-specific computation. Continuous Depth Batching (Schwethelm et al., 2026) exits early during decode; Position-Aware Depth Decay (Fan et al., 2025) skips more layers for later tokens; OverFill (Kim et al., 2025) decodes with a width-pruned model; Fixed Points Part II (Huang et al., 2026) distils a light prefill network; POP (He et al., 2026) and SPEED (Oh et al., 2026) prefill shallower than they decode;

Decode-Branch Transformers (Liu et al., 2026), PHD (Wu et al., 2025b) and State-Prediction Separation (Langford et al., 2026) add decode-side computation. We instead vary prefill and decode loops independently within one loop adapter and measure what each buys.

Asymmetric KV producers and consumers. YOCO (Sun et al., 2024) has a self-decoder produce one global KV cache that a cross-decoder reads; YOCO-U (Sun et al., 2026) loops the self-decoder, keeping KV storage independent of the loop count while every token still runs the loops; KITE (Hu et al., 2026) expands a checkpoint into two towers so that bulk prefill runs only the KV-producing tower. These architectures are built for the split; we adapt an existing looped model after pretraining and choose the prefill loops m and decode loops n per request.

Sharing and transferring state. Looped Latent Attention (Neill and Reid, 2026) compresses KV across loops; BeNice! (Prairie et al., 2026) and LSTM-style looped transformers (Kirsch, 2026) share it; Cache-to-Cache (Fu et al., 2025), HeteroFold (Yun et al., 2026), Swarm Telepathy (Alabdulkareem et al., 2026) and CacheLink (Qiu et al., 2026) transfer KV between models, and BellowsKV (Zhang and Chen, 2026) shares one latent across layers. Like CacheLink, we train our maps through the receiver's output; within one model tokens, layers and positions align, so a RoPE-commuting scale and a value map suffice. For hybrid attention / linear-recurrence models (Yang et al., 2024; Dong et al., 2024), state-swap interventions (Afendulev et al., 2026) and LatentPort (Villani, 2026) study the two kinds of state; Appendix H relates them to prefill loops.

3 Method

We describe the loop adapter for Ouro; SenseNova-Loop-9B uses a lighter variant without maps (Appendix A).

Prefill and decode. "prefill m / decode n" prefills the prompt with m loops and decodes every answer token with n loops; native n/n is the unmodified model. The frozen model prefills the prompt with m loops and keeps only the last loop's prompt keys and values. The decoder is the same frozen model with rank-16 LoRA (Hu et al., 2021) on its attention projections, running n loops per token.

RoPE-commuting maps. Decode loop k reads the prompt KV cache through a map per (loop, layer, KV head). The key map is one complex scale c_j per rotary frequency pair j (Su et al., 2021): viewing the pair of key coordinates (k_{2j}, k_{2j+1}) as a 2-D vector, $c_j = r\,e^{i\phi}$ stretches it by r and turns it by ϕ. Writing a key pair rotated for position s as a complex number $\tilde{k}_j = e^{is\theta_j} k_j$, scaling commutes with the rotation, $c_j \tilde{k}_j = e^{is\theta_j}(c_j k_j)$, and $\mathrm{Re}\langle q_j, c_j \tilde{k}_j\rangle = \mathrm{Re}\langle \bar{c}_j q_j, \tilde{k}_j\rangle$: the map is applied to the query and the stored cache is never rewritten. The value map is a full linear map W_k per head; because attention is linear in the values, $\sum_i a_i W_k v_i = W_k \sum_i a_i v_i$ is applied once to the attention output over the prompt. Prompt and generated tokens share one softmax: the two parts are computed by two attention calls with outputs o_P, o_A and log-sum-exps λ_P, λ_A of their attention logits, and merged as $g\,o_P + (1-g)\,o_A$ with $g = \sigma(\lambda_P - \lambda_A)$, which equals one softmax over both (the blockwise softmax of FlashAttention (Dao, 2023)). The prompt KV is therefore stored once for every n, and total KV is $b(\ell + n\,t)$ instead of $b\,n(\ell + t)$. Serving from one cache equals materialising a mapped prompt cache per loop; in bfloat16 all 512 checked greedy tokens agree.

What the maps do. With the same training and the maps frozen at the identity, the decoder loses little on short-context reasoning and knowledge but 11.4 points on RULER-16K at prefill 4 / decode 1 (Appendix F): the maps matter mainly for long prompts.

Training. We minimise $\mathrm{KL}(\text{teacher} \,\|\, \text{student})$ over answer positions, with the native 4/4 model as teacher, on 48M answer tokens of reasoning traces, instruction data, long-context data and web text. One (m, n) pair is sampled per micro-batch, so one set of parameters (63M, 2.4% of Ouro-2.6B) serves all 16 pairs. Details are in Appendix A.

4 Experiments

Setup. Models: Ouro-2.6B, Ouro-1.4B and their Thinking variants (Zhu et al., 2025) (four loops), Nanbeige4.2-3B-Base (two loops) and SenseNova-Loop-9B-Preview, a hybrid of Gated-DeltaNet and full-attention layers whose middle block is visited twice. Benchmarks: GSM8K (Cobbe et al., 2021), MATH-500 (Hendrycks et al., 2021; Lightman et al.,

```
# blocks: frozen looped block, L layers; m/n: prefill/decode loops
# c[k][l] = 1+a+1j*b per RoPE pair; W[k][l] = I+delta (init: I)
# LoRA on q, k, v, o is active only while decoding.

def prefill(prompt, m): # prompt minus its last token
    h = embed(prompt)
    for _ in range(m):
        for l in range(L):
            h, K[l], V[l] = blocks[l](h) # last loop's K, V kept
    return K, V # one prompt KV cache

def attend(q, K, V, kv_answer, c, W): # layer l, decode loop k
    qc = complex_mul(q, c.conj()) # key scale moved to the query
    o_p, lse_p = flash_attn(qc, K, V) # prompt part
    o_p = o_p @ W.T # attention is linear in V
    o_a, lse_a = flash_attn(q, *kv_answer, causal=True)
    g = torch.sigmoid(lse_p - lse_a) # exact softmax over both
    return g * o_p + (1 - g) * o_a

def decode(token, K, V, n): # last prompt token, then answer
    h = embed(token)
    for k in range(n):
        for l in range(L):
            h = blocks[l](h, attn=lambda q, kv: attend(
                    q, K[l], V[l], kv, c[k][l], W[k][l]))
    return lm_head(h)
```

```
# training: one (m, n) per micro-batch,
# teacher = the native 4/4 model
m, n = randint(1, 4), randint(1, 4)
with torch.no_grad(): t = model(x, prefill=4, decode=4)
s = model(x, prefill=m, decode=n) # LoRA + maps
kl = kl_div(s.log_softmax(-1),
          t.log_softmax(-1), log_target=True,
          reduction="none").sum(-1)
loss = kl[answer].sum() / answer.sum()
```

Figure 3: **The loop adapter in PyTorch-style pseudocode.** The prompt KV cache is written once by the last of m prefill loops; every decode loop reads it through its own key scale (applied to the query) and value map (applied to the attention output), and the prompt and answer attention are combined exactly through their log-sum-exp.

2023), MMLU (Hendrycks et al., 2020), MMLU-Pro (Wang et al., 2024), ARC-Challenge (Clark et al., 2018), BBH (Suzgun et al., 2022), RULER (Hsieh et al., 2024) at 4K/8K/16K and HELMET (Yen et al., 2024); protocols and set sizes are in Appendix B. Every comparison is paired over items with 95% bootstrap intervals. To isolate prefill loops from adaptation we compare prefill m / decode n with prefill n / decode n of the *same* loop adapter; comparisons with native models include the effect of training the loop adapter.

4.1 What prefill loops and decode loops buy

Figure 1 shows the full grid on Ouro-2.6B. MMLU rises from 52–53 at one prefill loop to 70 at three or four with one decode loop and to 74–75 with more decode loops; GSM8K gains 9 points along the prefill axis at one decode loop but rises from 46–55 to 78–88 with decode loops. Table 2 confirms the pattern on full sets: at one decode loop, prefilling with four loops instead of one adds 15.2 MMLU, 8.6 RULER-16K and 8.0 GSM8K points; the gain shrinks as decode loops are added. A second training seed reproduces these numbers (Appendix C).

The same holds across models (Table 1): with one decode loop, more prefill loops improve knowledge and long-context scores on all five checkpoints (the Ouro-1.4B RULER-16K gain is not significant). The effect depends on the task and its prompt; Appendix J lists where more prefill loops hurt, notably few-shot MATH-500 and 3-shot BBH on the 1.4B models.

4.2 More prefill loops, fewer decode loops

Figure 4 puts the two findings together on Ouro-2.6B-Thinking: prefill 4 / decode 2. At the decode cost of native 2/2, prefill 4 / decode 2 scores higher on all eight benchmarks, significantly on MMLU-Pro ($+6.9$ [3.1, 11.0]), BBH ($+6.9$ [4.6, 9.2]), ARC-Challenge ($+3.6$ [2.0, 5.3]), MBPP+ ($+4.8$ [1.3, 8.2]) and HELMET ($+13.0$ [6.7, 19.5]); on GSM8K, MATH-500 ($+0.2$ [-2.6, 3.0]) and HumanEval the gains are within noise. Against native 4/4, which decodes with twice the loops, it is within two points on six of the eight benchmarks and trails on HELMET (-8.7), where native 4/4 recalls exact key–value pairs that more prefill loops lose (Appendix J), and on MATH-500 (-4.8 [-7.8, -2.0]).

Part of the gain over native 2/2 comes from training the loop adapter, not from the loop counts. The same loop adapter prefilling with two loops (hollow circles; same weights, same training, same decode loops) isolates prefill loops: prefilling with four rather than two loops adds $+8.1$ [4.0, 12.1] on MMLU-Pro, $+7.5$ [1.4, 13.8] on HELMET, $+2.3$ [0.9, 3.7] on ARC-Challenge and $+2.2$ [0.1, 4.3] on BBH, and shows no detectable difference on GSM8K ($+0.2$), MATH-500 ($+0.2$)

model (same decoder, decode 1)	knowledge	long context	multi-step
Ouro-2.6B, prefill 4 vs. 1	MMLU +15.2 [14.3, 16.1]	RULER-16K +8.6 [4.3, 12.8]	GSM8K +8.0 [5.3, 10.5]
Ouro-1.4B, prefill 4 vs. 1	ARC-C +10.8 [8.4, 13.1]	RULER-16K +1.4 [−2.6, 5.4]	GSM8K +6.2 [3.6, 8.7]
Ouro-2.6B-Thinking, prefill 4 vs. 1	MMLU-Pro +18.6 [13.6, 23.8]	HELMET +7.7 [2.1, 13.6]	GSM8K +6.6 [4.2, 9.2]
Ouro-1.4B-Thinking, prefill 4 vs. 1	MMLU-Pro +17.1 [12.4, 22.4]	HELMET +4.6 [0.1, 8.6]	GSM8K +13.6 [10.5, 16.6]
SenseNova-Loop-9B, prefill 2 vs. 1	MMLU-Pro +5.5 [2.9, 8.1]	HELMET +3.5 [0.8, 6.1]	BBH +7.0 [5.9, 8.0]

Table 1: **Prefill loops with the decoder held fixed** across five checkpoints (paired, within one loop adapter). Set sizes: MMLU 14,042, ARC-C 1,172, MMLU-Pro 420 (Ouro) and 980 (SenseNova), RULER 520, HELMET 1,108 (Ouro) and 6,646 (SenseNova), GSM8K 1,319, BBH 6,511. HELMET is averaged as its authors do (mean per dataset, then per category, then over its four categories). Each column shows one benchmark of that kind per checkpoint; counterexamples are discussed in Section 4.1 and Appendix J. Differences are score points on a 0–100 scale; brackets are paired 95% bootstrap intervals.

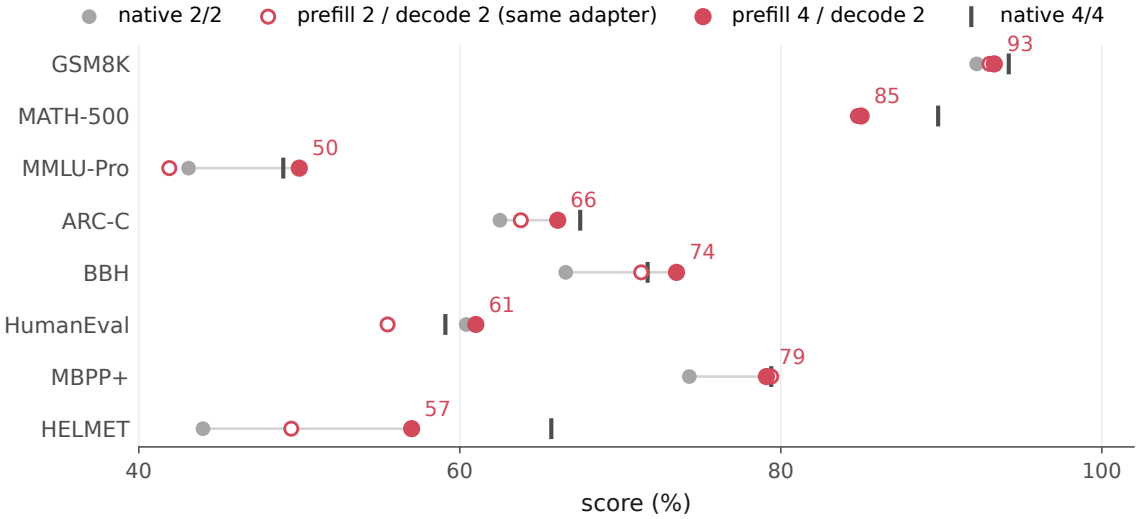


Figure 4: **Prefill 4 / decode 2: the decode cost of two loops, close to the quality of four.** Ouro-2.6B-Thinking on standard benchmarks (GSM8K, MATH-500, MMLU-Pro, ARC-Challenge, BBH, HumanEval, MBPP+, and HELMET averaged over its categories); prefill 4 / decode 2 uses one loop adapter and one stored prompt KV cache. Hollow circles: the same loop adapter prefilling with only two loops, which separates the effect of training from that of prefill loops. All models are scored on identical evaluation sets; Table 4 gives every score and paired interval.

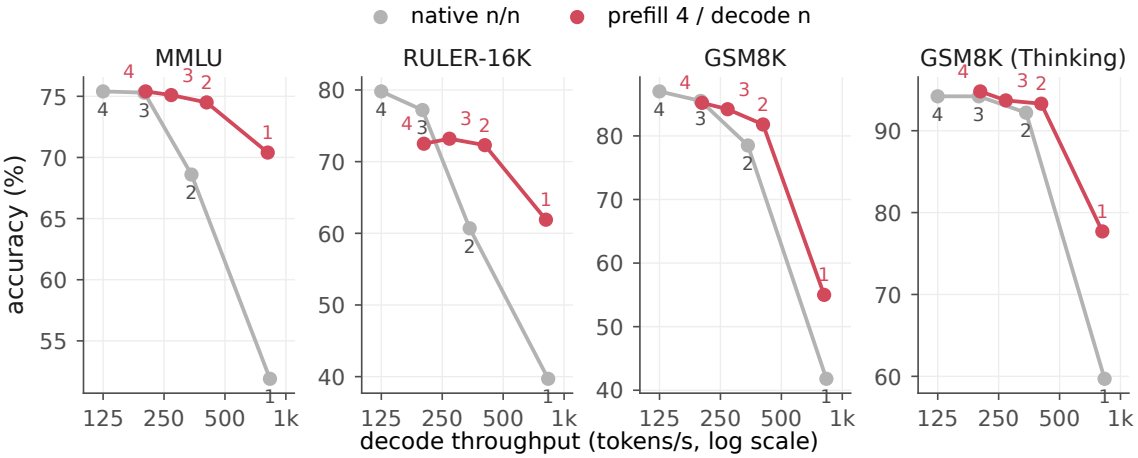


Figure 5: **Accuracy against decode throughput.** Native n/n (grey) and the loop adapter with prefill 4 / decode n (red), $n = 1$–4 (labels). Throughput: best decode tokens/s over batch sizes at 8K context on one B300 (same for the base and Thinking models, which share the architecture); one stored prompt KV cache admits larger batches as n grows. Ouro-2.6B on the grid subsets of Figure 1; Ouro-2.6B-Thinking on all 1,319 GSM8K problems.

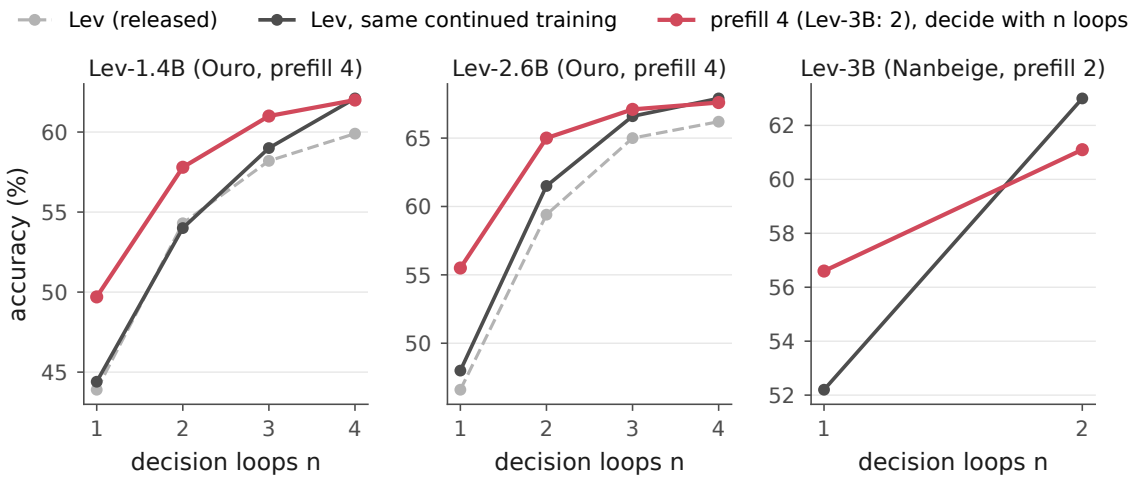


Figure 6: **More prefill loops, fewer decision loops.** Lev looped decision models (Song et al., 2026) on the public LevBench development set (3,270 answerable questions): accuracy against the number of loops spent on the decision. Red: the context is prefilled with four loops (two on Nanbeige) and the question and options run n loops on that KV; dark grey: the same continued training with Lev's own forward; dashed: the released model.

decode n	MMLU	RULER-16K	GSM8K
1	$+15.2$ [14.3, 16.1]	$+8.6$ [4.3, 12.8]	$+8.0$ [5.3, 10.5]
2	$+6.0$ [5.4, 6.7]	$+5.3$ [1.3, 9.4]	-1.7 [-3.6, 0.2]
3	$+0.6$ [0.2, 1.1]	-0.4 [-3.0, 2.2]	-0.8 [-2.4, 0.8]

Table 2: **Prefill loops at fixed decode loops** (Ouro-2.6B, one loop adapter, full sets): prefill 4 / decode d minus prefill n / decode n. Differences are score points on a 0–100 scale; brackets are paired 95% bootstrap intervals.

and MBPP+ (-0.3): the gains over native 2/2 on knowledge and long-context benchmarks track prefill loops, the others do not. The same holds on Ouro-1.4B-Thinking (Appendix D).

Figure 5 shows the same trade-off for every number of decode loops, against measured decode throughput. For every n, prefilling with four loops moves the model up (more accurate at one or two decode loops: MMLU $+18.4$ and $+5.9$, RULER-16K $+22.2$ and $+11.6$ on the grid subsets) or right (faster at three or four decode loops, because one stored prompt cache admits larger batches: 271 and 203 against 200 and 125 tokens/s). Prefill 4 / decode 2 runs at $1.19\times$ the throughput of native 2/2 and $3.2\times$ that of native 4/4.

4.3 Fewer decode loops are faster end to end

We time whole answers at batch 1, prefill included (Figure 7; Table 5 in Appendix E). On Ouro-2.6B-Thinking a loop adapter that writes with two loops answers GSM8K in $0.55\times$ [0.45, 0.67] the time of native 4/4. On all 1,319 problems, prefill 4 / decode 2 scores 93.3 against 94.2, a paired difference of -1.0 [-2.0, 0.1]. Prefilling with 4 rather than 2 loops before decoding with two changes the answer time by $1.01\times$ [0.90, 1.11]: the saving comes from the adapted decoder with fewer loops, including its shorter answers. On MATH-500, whose answers it does not shorten, the saving is smaller: $0.63\times$ the time of native 4/4 but $1.22\times$ that of native 2/2 (Appendix E). On SenseNova-Loop-9B the trained prefill 1 / decode 1 scores 95.6 against 95.8 for native 2/2 on all 1,319 GSM8K problems and takes $0.59\times$ [0.52, 0.68] the time per answer, close to the ratio of layer executions per token (32 vs. 56).

4.4 Decision models

The split also serves looped decision models. Lev (Song et al., 2026) reads a context, then scores the options of a question with a learned head after every loop, without generating text. We prefill the context with four loops (two for Lev-3B, whose backbone has two), let every decision loop attend to the last prefill loop's KV, and continue training Lev's own LoRA and scoring head; the control continues the same weights on the same data and number of updates with Lev's own forward. We evaluate on the 3,270 answerable questions of LevBench's public development set (never its locked test set), with intervals over record groups as Lev does; setup and further checks are in Appendix G.

With one decision loop, prefilling with more loops raises accuracy by $+5.3$ [3.4, 7.2] on Lev-1.4B, $+7.5$ [5.7, 9.5] on Lev-2.6B and $+4.4$ [2.8, 6.1] on Lev-3B (Figure 6); the advantage shrinks as de-

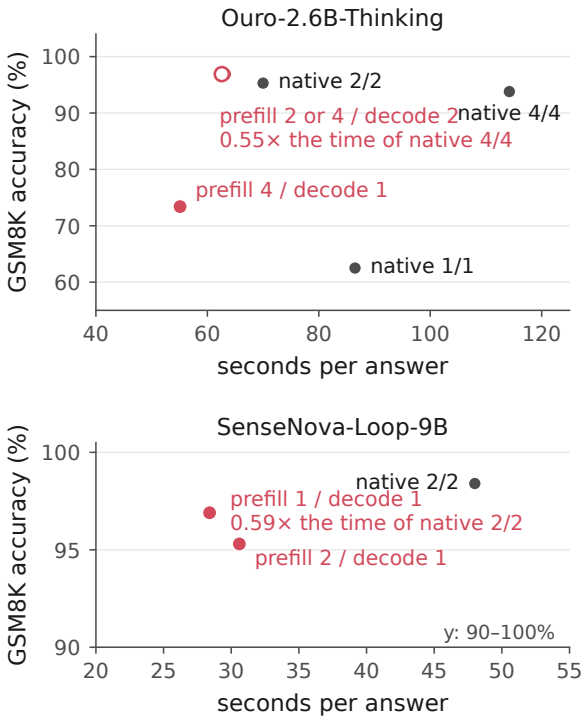


Figure 7: **Fewer decode loops save time end to end.** Accuracy vs. seconds per answer at batch 1 (prefill included, real greedy answers, the same 64 GSM8K problems for every point). Top: decoding with two loops takes $0.55\times$ the time of native 4/4, whether the prompt is prefilled with 2 or 4 loops; on all 1,319 problems prefill 4 / decode 2 is 1.0 point below native 4/4 (Section 4.3). Bottom: on SenseNova-Loop-9B, whose native one-loop model solves 2.2% of all GSM8K problems, prefill 1 / decode 1 is 1.6 points below native 2/2 at $0.59\times$ the time (y axis zoomed).

cision loops are added; with all of them it is gone on the Ouro backbones and negative on Lev-3B (-1.8). It holds across training seeds (four per backbone; all significant except one Lev-3B seed) and against a control trained for twice as many updates ($+4.0$, $+5.5$, $+4.0$), so it is not explained by the extra computation of our training forward.

5 Conclusion

A looped language model need not run the same number of loops for prefill and decode. On the checkpoints and benchmarks we test, prefill loops mainly buy knowledge and long-context accuracy and decode loops mainly buy multi-step math, so the best pair depends on the task. A small loop adapter unties the two on existing checkpoints with one exact prompt cache; with more prefill than decode loops it reaches near-native quality at a lower decode cost and answer time, and the same untying helps looped decision models.

Limitations

More prefill loops lower 4-shot MATH-500, give a negative but inconclusive estimate on RULER-4K and, on Ouro, lower exact key–value recall in HELMET (Tables 11 and 10); at three or more decode loops, one prompt KV cache from the fourth loop trails the native model on RULER ($-10.4/-9.5/-5.3$ at 4K/8K/16K, decode 3). More prefill loops raise time to first token, so it pays when answers are long or quality requires it. Both 1.4B checkpoints degenerate at prefill 4 / decode 1 on 3-shot BBH (Ouro-1.4B: -24.5, Ouro-1.4B-Thinking: -20.5 against prefill 1 / decode 1); with two decode loops, prefilling with four instead of two helps (Ouro-1.4B-Thinking: $+8.8$, Table 4). On Nanbeige4.2-3B-Base our shared-loop adapter recipes trail loop adapters trained per (m, n) pair, so Appendix B reports per-pair results. Gains against native models include the adaptation of the loop adapter. Throughput is a controlled decode benchmark and end-to-end time is measured at batch 1 on one GPU type. Thinking checkpoints on raw HELMET prompts can stop at once; we decode them so that an answer cannot be empty, for every configuration (Appendix B). Probes measure linear decodability, not causal use. All models are 1.4B–9B, decoding is greedy and benchmarks are English.

References

Kirill Afendulev, Alexey Dontsov, Elena Tutubalina, and Anton Korznikov. 2026. What Attention Recalls and Recurrence Controls in Hybrid Language Models. *arXiv preprint arXiv:2609.04434*.

Abdulrahman Alabdulkareem, Irene Tenison, Yang Zhang, Yada Zhu, Boris Katz, and Dan Gutfreund. 2026. Swarm Telepathy: Prefill-Free KV-Cache Transfer Across Families. *arXiv preprint arXiv:2610.15095*.

Sangmin Bae, Yujin Kim, Reza Bayat, Sungnyun Kim, Jiyoun Ha, Tal Schuster, Adam Fisch, Hrayr Harutyunyan, Ziwei Ji, Aaron Courville, and Se-Young Yun. 2025. Mixture-of-Recursions: Learning Dynamic Recursive Depths for Adaptive Token-Level Computation. *arXiv preprint arXiv:2507.10524*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. *arXiv preprint arXiv:1803.05457*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias

Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168*.

Tri Dao. 2023. FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning. *arXiv preprint arXiv:2307.08691*.

Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. 2018. Universal Transformers. *arXiv preprint arXiv:1807.03819*.

Xin Dong, Yonggan Fu, Shizhe Diao, Wonmin Byeon, Zijia Chen, Ameya Sunil Mahabaleshwarkar, Shih-Yang Liu, Matthijs Van Keirsbilck, Min-Hung Chen, Yoshi Suhara, Yingyan Lin, Jan Kautz, and Pavlo Molchanov. 2024. Hymba: A Hybrid-head Architecture for Small Language Models. *arXiv preprint arXiv:2411.13676*.

Siqi Fan, Xuezhi Fang, Xingrun Xing, Peng Han, Shuo Shang, and Yequan Wang. 2025. Position-Aware Depth Decay Decoding (D^3): Boosting Large Language Model Inference Efficiency. *arXiv preprint arXiv:2503.08524*.

Tianyu Fu, Zihan Min, Hanling Zhang, Jichao Yan, Guohao Dai, Wanli Ouyang, and Yu Wang. 2025. Cache-to-Cache: Direct Semantic Communication Between Large Language Models. *arXiv preprint arXiv:2510.03215*.

Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. 2025. Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach. *arXiv preprint arXiv:2502.05171*.

Junhui He, Zhihui Fu, Jun Wang, and Qingan Li. 2026. POP: Prefill-Only Pruning for Efficient Large Model Inference. *arXiv preprint arXiv:2602.03295*.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2020. Measuring Massive Multitask Language Understanding. *arXiv preprint arXiv:2009.03300*.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring Mathematical Problem Solving With the MATH Dataset. *arXiv preprint arXiv:2103.03874*.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024. RULER: What's the Real Context Size of Your Long-Context Language Models? *arXiv preprint arXiv:2404.06654*.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.

Zhiheng Hu, Yixun Wei, Jian Zhou, Yizhuang Zhou, Ji Li, Xing Chen, Yang Li, Bojun Wang, Yibo Zhu, Xiangyu Zhang, and Daxin Jiang. 2026. KITE: KV-Invariant Transformer Expansion for Efficient Agentic LLM Scaling. *arXiv preprint arXiv:2609.27294*.

Benhao Huang, Chufan Shi, Junlin Chen, Shicheng Wen, Zhengzhong Liu, Eric Xing, and Xuezhe Ma. 2026. Towards Looped Models Done Right, Part II: Rethinking at Fixed Points. *arXiv preprint arXiv:2610.06833*.

Woojeong Kim, Junxiong Wang, Jing Nathan Yan, Mohamed Abdelfattah, and Alexander M. Rush. 2025. OverFill: Two-Stage Models for Efficient Language Model Decoding. *arXiv preprint arXiv:2508.08446*.

Andreas Kirsch. 2026. LSTM-Style Looped Transformers with Shared KV Caches. *arXiv preprint arXiv:2610.33360*.

Faisal Ladhak, Shantanu Acharya, Santiago Akle Serano, Simeng Sun, Boris Ginsburg, and Krishna C Puvvada. 2026. One Model to Loop Them All: Elastic Latent Recurrence for Language Models. *arXiv preprint arXiv:2610.12582*.

John Langford, Nathan Godey, Giovanni Monea, Yoav Artzi, Harry Dong, Ying Fan, Gustavo de Rosa, and Zheng Zhan. 2026. Almost Free State Prediction Separation. *arXiv preprint arXiv:2609.03807*.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let's Verify Step by Step. *arXiv preprint arXiv:2305.20050*.

Liming Liu, Mingze Wang, and Tuo Zhao. 2026. Decode-Branch Transformers: Decoupling the Primary Prefill Path from Additional Decode Computation. *arXiv preprint arXiv:2608.12385*.

Aur Shalev Merin. 2026. Present but Hard to Read: Probing Latent Reasoning with Minimal Pairs. *arXiv preprint arXiv:2610.28498*.

Giovanni Monea, Keshav Ramji, Yousef El-Kurdi, Luis A. Lastras, Yoav Artzi, Nathan Godey, and Ramón Fernandez Astudillo. 2026. The Surprising Effectiveness of Shared Memory in Looped Transformers. *arXiv preprint arXiv:2610.02383*.

James O' Neill and Fergal Reid. 2026. Looped Latent Attention: Cross-Loop KV Compression for Looped Transformers. *arXiv preprint arXiv:2607.15456*.

Jungsuk Oh, Hyeseo Jeon, Hyunjune Ji, Kyongmin Kong, and Jay-Yoon Lee. 2026. Shallow Prefill, Deep Decoding: Efficient Long-Context Inference via Layer-Asymmetric KV Visibility. *arXiv preprint arXiv:2605.06105*.

Hayden Prairie, Jacob M. Ryan, Taylor Berg-Kirkpatrick, and Daniel Y. Fu. 2026. BeNice! Closing Looping's Inference-Memory Gap via KV-Sharing. *arXiv preprint arXiv:2610.31633*.

Haomiao Qiu, Yining Tang, Zijia Song, Ruizi Han, Ziyue Qiao, and Miao Zhang. 2026. CacheLink: Functional KV Transport for Zero-Prefill Cross-Model Context Handoff. *arXiv preprint arXiv:2610.15509*.

Kristian Schwethelm, Daniel Rueckert, and Georgios Kaissis. 2026. Depth-adaptive Inference of Looped Language Models via Continuous Depth Batching. *arXiv preprint arXiv:2608.09444*.

Dingjie Song, Zhiling Yan, Yuxuan Zhang, and Lichao Sun. 2026. LEV: Thinking Fast and Deep with Looped Decision Models. *arXiv preprint arXiv:2610.20951*.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. 2021. RoFormer: Enhanced Transformer with Rotary Position Embedding. *arXiv preprint arXiv:2104.09864*.

Yutao Sun, Li Dong, Tianzhu Ye, Shaohan Huang, Jianyong Wang, and Furu Wei. 2026. Universal YOCO for Efficient Depth Scaling. *arXiv preprint arXiv:2604.01220*.

Yutao Sun, Li Dong, Yi Zhu, Shaohan Huang, Wenhui Wang, Shuming Ma, Quanlu Zhang, Jianyong Wang, and Furu Wei. 2024. You Only Cache Once: Decoder-Decoder Architectures for Language Models. *arXiv preprint arXiv:2405.05254*.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. 2022. Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them. *arXiv preprint arXiv:2210.09261*.

Simon P. Villani. 2026. LatentPort: Beyond KV Cache - Cross-Model Transfer of Recurrent Memory in Hybrid Language Models: A 4B-to-9B Hybrid-State Handoff Without Target Prefix Replay. *arXiv preprint arXiv:2609.25053*.

Elena Voita, Rico Sennrich, and Ivan Titov. 2019. The Bottom-up Evolution of Representations in the Transformer: A Study with Machine Translation and Language Modeling Objectives. In *Proceedings of EMNLP-IJCNLP 2019*, pages 4396–4406.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. 2024. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. *arXiv preprint arXiv:2406.01574*.

Bohong Wu, Mengzhao Chen, Xiang Luo, Shen Yan, Qifan Yu, Fan Xia, Tianqi Zhang, Hongrui Zhan, Zheng Zhong, Xun Zhou, Siyuan Qiao, and Xingyan Bin. 2025a. Parallel Loop Transformer for Efficient Test-Time Computation Scaling. *arXiv preprint arXiv:2510.24824*.

Bohong Wu, Shen Yan, Sijun Zhang, Jianqiao Lu, Yutao Zeng, Ya Wang, and Xun Zhou. 2025b. Efficient Pretraining Length Scaling. *arXiv preprint arXiv:2504.14992*.

Jian Yang, Shawn Guo, Wei Zhang, Tianyu Zheng, Yaxin Du, Haau-Sing Li, Jiajun Wu, Yue Song, Yan Xing, Qingsong Cai, Zelong Huang, Chuan Hao, Ran Tao, Xianglong Liu, Wayne Xin Zhao, Mingjie Tang, Weifeng Lv, Ming Zhou, and Bryan Dai. 2026. LoopCoder-v2: Only Loop Once for Efficient Test-Time Computation Scaling. *arXiv preprint arXiv:2606.18023*.

Liu Yang, Kangwook Lee, Robert Nowak, and Dimitris Papailiopoulos. 2023. Looped Transformers are Better at Learning Learning Algorithms. *arXiv preprint arXiv:2311.12424*.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. 2024. Gated Delta Networks: Improving Mamba2 with Delta Rule. *arXiv preprint arXiv:2412.06464*.

Howard Yen, Tianyu Gao, Minmin Hou, Ke Ding, Daniel Fleischer, Peter Izsak, Moshe Wasserblat, and Danqi Chen. 2024. HELMET: How to Evaluate Long-Context Language Models Effectively and Thoroughly. *arXiv preprint arXiv:2410.02694*.

Vincent-Daniel Yun, Woosang Lim, Haneul Yoo, Sungjoo Yoo, Murali Annavaram, and Sai Praneeth Karimireddy. 2026. Prefill-Free Cross-Family KV Cache Transfer for Heterogeneous Multi-Agent LLMs. *arXiv preprint arXiv:2609.32259*.

Jiamu Zhang and Hanjie Chen. 2026. BellowsKV: One Latent for Many Layers, Expanded on Demand. *arXiv preprint arXiv:2610.35023*.

Yudong Zhang, Wei Ding, Kaili Zheng, Yuchen Ding, and Zhenyu Zhang. 2026. Accordion: Elastic-Depth Inference via Fixed-Prefix Layer Looping. *arXiv preprint arXiv:2610.32492*.

Rui-Jie Zhu, Zixuan Wang, Kai Hua, Tianyu Zhang, Ziniu Li, Haoran Que, Boyi Wei, Zixin Wen, Fan Yin, He Xing, Lu Li, Jiajun Shi, Kaijing Ma, Shanda Li, Taylor Kergan, Andrew Smith, Xingwei Qu, Mude Hui, Bohong Wu, and 14 others. 2025. Scaling Latent Reasoning via Looped Language Models. *arXiv preprint arXiv:2510.25741*.

A Training Details

Loop adapter. Rank-16 LoRA ($\alpha = 16$, no dropout) on the query, key, value and output projections, active only when decoding; per (decode loop, layer, KV head) a complex scale for each rotary frequency pair on keys and a full $D \times D$ value map, both initialised to the identity. Trainable parameters: 63,307,776 for Ouro-2.6B. The backbone is frozen.

Objective and data. $\mathrm{KL}(\text{teacher} \,\|\, \text{student})$ summed over answer positions and normalised by the answer tokens in the global batch; teacher = native 4/4 with the same weights; one pair (m, n) sampled per micro-batch. 48M answer tokens: OpenThoughts reasoning traces 45%, Tulu SFT responses 40%, LongAlpaca long-context answers 5% and FineWeb-Edu windows 10% (fractions of answer tokens); prompts up to 7,680 tokens, answers up to 1,024. Base models see a plain "User: … Assistant:" format; thinking models see the same samples in their chat template. No benchmark data was deliberately included; we did not decontaminate the training corpora.

Optimisation. AdamW ($\beta = (0.9, 0.95)$, no weight decay), peak learning rate 10^{-4} with 2% linear warm-up and cosine decay to 10^{-5}, gradient clipping at 1.0, global batch 24 sequences on 4 B300 GPUs (6.75 hours for Ouro-2.6B), bfloat16 backbone.

Answer stored once. Generated tokens keep one KV cache per layer shared by all decode loops; at inference the last decode loop commits each token, and earlier loops read that history plus the current token's KV computed live in their own loop (*self-live*[1], a layout also used by LSTM-style looped transformers (Kirsch, 2026)). For example, when loop 1 processes token 5, it attends to tokens 1–4 through the last loop's stored KV and to token 5 through its own, freshly computed KV. Training uses the same KL objective in this layout, in two passes: a no-grad pass in which each loop reads its own causal answer KV records the last loop's answer KV; a second pass reads, at every loop, the prompt KV cache, that recorded history strictly before the current position, and the current token's KV computed live on the diagonal. Training thus approximates, but does not exactly reproduce, inference, where the history is generated recursively in the shared cache. With prompt and answer each stored once, total KV is $b(n+t)$, that of a one-loop model; with 1K-token prompts and 2,048 generated tokens the largest batch that fits grows from 32 (native 4/4) to 192 and decode-4 throughput from 219 to 476 tokens/s (759 vs. 586 at n=2), at a cost of 4.8 GSM8K points against native 4/4 (Table 7 compares it with per-loop answer caches). On Ouro-2.6B-Thinking, a loop adapter trained the same way changes full-set GSM8K at prefill 4 / decode 2 by $+0.5$ [−0.5, 1.6] against per-loop answer caches; at batch 1 its cost per generated token is the same (82.5 vs. 83.1 ms) but its answers are longer (1,166 vs. 741 tokens), so it takes $1.53\times$ [1.27, 1.84] the time per answer. Its benefit is memory and batch size, not batch-1 latency.

SenseNova-Loop-9B. Rank-16 LoRA on the attention projections of the looped layers (full attention: q/k/v/o; Gated-DeltaNet: input and output projections), active only when decoding; no maps. The one-visit decoder reads the prefill state of the last of $m \in \{1, 2\}$ visits (sampled 50/50), so the same decoder serves prefill 1 and prefill 2. The untrained condition passes the last visit's full-attention KV and Gated-DeltaNet convolution and recurrent states. Teacher: native 2/2; 20M answer tokens of the same mixture in the SenseNova chat template; peak learning rate 2×10^{-4}.

B Evaluation Details

Base models use lm-evaluation-harness prompts through our runner for GSM8K (3-shot CoT, flexible-extract, 1,319), MMLU (5-shot, 14,042), ARC-Challenge (25-shot, 1,172), BBH (3-shot CoT, 60 per subtask) and MATH-500 (the 500-problem subset of MATH, 4-shot Minerva prompt). RULER (13 tasks × 40 rows per length) and HELMET (RAG, synthetic recall, re-ranking and ICL categories; 1,108 rows) are prepared and scored by our own scripts. Thinking models answer GSM8K and MATH-500 zero-shot in their chat template, answers in `\boxed{}` scored with math-verify, with budgets of 4,096 and 8,192 tokens; their MMLU-Pro is 5-shot CoT with 30 questions per subject (420) and their BBH 3-shot CoT with 60 per subtask. SenseNova-Loop-9B uses MMLU-Pro with 70 questions per subject (980) through the harness and the full BBH (6,511) through our own scripts. For base models, MATH-500 answers are scored with math-verify on the full generated answer, with and without exemplars. The harness's own math-verify extraction gives -9.8 instead of -11.2 for the 4-shot comparison on Ouro-2.6B; Minerva's strict "Final Answer" match would reverse it (19.6 vs. 9.8) because the prefill-1 decoder drops the required format more often. On raw HELMET prompts thinking checkpoints can emit a newline and stop; for thinking models we therefore mask the stop tokens until an answer has a non-whitespace token, for

[1] Self-live was developed by the author in separate work on shared KV caches for looped models; here it is applied to the connector's answer tokens.

every configuration. SenseNova RULER uses a non-thinking prefill for every configuration. Native throughput baselines: static per-loop caches with the same two-call flash attention (fastest), the standard dynamic cache (500 / 245 / 105 tokens/s at $n = 1/2/4$), and a preallocated contiguous cache with one attention call (267 / 130 / 57); all produce identical greedy tokens. Nanbeige4.2-3B-Base loop adapters trained per (m, n) pair, prefill 2 / decode 1 minus prefill 1 / decode 1: RULER 4K/8K/16K $+11.8/+11.7/+14.2$, MMLU $+12.2$, GSM8K $+5.2$. Intervals: paired bootstrap over items, 5,000 resamples; every interval in the paper is recomputed from released per-item scores by one script (Appendix L).

C Training-Seed Robustness

	task	seed 1	seed 2
prefill 4 − prefill 1 (decode 1)	GSM8K	$+8.0$	$+7.2$
	MMLU	$+15.2$	$+15.2$
	RULER-4K	-3.8	-3.5
	RULER-8K	$+2.5$	$+2.0$
	RULER-16K	$+8.6$	$+8.6$
prefill 4 − 2 (decode 2)	GSM8K	-1.7	-2.9
	RULER-4K	-7.2	-6.8

Table 3: Ouro-2.6B loop adapter trained with two seeds (initialisation and (m, n) sampling); paired differences within each seed, full sets.

D Prefill 4 / Decode 2 in Numbers, and on Ouro-1.4B-Thinking

Table 4 lists the scores and paired intervals behind Figure 4 and repeats the comparison on the smaller Thinking checkpoint. The effect of prefill loops replicates: with the same loop adapter, prefilling with four loops instead of two raises MMLU-Pro ($+8.6$), HELMET ($+7.3$), ARC-Challenge ($+2.7$) and BBH ($+8.8$) and shows no detectable difference on GSM8K, MATH-500, HumanEval and MBPP+. Against native 2/2 the result is weaker than on Ouro-2.6B-Thinking: prefill 4 / decode 2 is significantly better on GSM8K ($+3.0$) and ARC-Challenge ($+4.3$), within noise on five benchmarks, and much worse on 3-shot BBH (-15.9), where this checkpoint's loop adapter degenerates at both prefill loop counts (Limitations). Against native 4/4 the gap is larger than on Ouro-2.6B-Thinking, most of all on BBH (-37.2) and HELMET (-18.3).

E End-to-End Time and Decode Throughput

Serving. Storing the prompt once also changes throughput (Figure 5; Table 6 in Appendix E). In a controlled decode benchmark (8K context, 32 repeated decode steps, largest batch that fits on one B300), one prompt KV cache decodes $0.97\times$ as fast as the fastest native implementation we measured at one decode loop and 1.19–$1.62\times$ at two to four, because it admits larger batches as n grows; this native baseline uses the same flash attention and is faster than the standard dynamic cache (Appendix B). With long answers (1K-token prompts, 2,048 generated tokens) one prompt KV cache decodes $0.97\times$ native at n=2 and $1.16\times$ at n=4.

F Design Variants

Table 7 compares design variants. Training the same loop adapter with its maps frozen at the identity, so that a LoRA decoder reads the last prefill loop's KV directly, gives small estimated losses on GSM8K and MMLU, with intervals including zero, but lowers RULER-16K by 11.4 points at prefill 4 / decode 1 and by 2.6–3.4 points at two decode loops. Restricting the key map to a RoPE-commuting scale, which is what allows one stored prompt KV cache, shows no measurable cost against unrestricted key maps with one cache per loop (RULER-16K, five (m, n) pairs).

G Decision Models: Setup and Checks

Both arms continue each released Lev model for 140 updates on the 8,925-record public S31 training subset with Lev's hyperparameters, from the same weights. Our reference forward runs the full sequence through both the prefill and the decision loops, so its training computation is about twice the control's. Without training, more prefill loops do not help (-3.0 on Lev-1.4B at one decision loop), and the control itself moves little against the released models at one loop ($+0.5$ on Lev-1.4B, $+1.3$ on Lev-2.6B). Repeating both arms with other seeds, the one-loop gain ranges from $+3.9$ to $+5.3$ on Lev-1.4B and from $+5.1$ to $+7.5$ on Lev-2.6B, every interval excluding zero (four seeds each); on Lev-3B it is $+4.4$, $+5.1$, $+4.0$ and $+0.9$ [-0.8, 2.7] (mean $+3.6$). A control trained for two epochs (twice the updates, about the training computation of the other arm) still trails it at one decision loop by $+4.0$ [2.1, 6.0] on Lev-1.4B,

	score				prefill 4 / decode 2 minus		
benchmark	native 2/2	prefill 2 / decode 2	prefill 4 / decode 2	native 4/4	native 2/2	prefill 2 / decode 2	native 4/4
Ouro-2.6B-Thinking (Figure 4)							
GSM8K	92.2	93.0	93.3	94.2	+1.1 [−0.2, 2.4]	+0.2 [−1.0, 1.4]	−1.0 [−2.0, 0.1]
MATH-500	84.8	84.8	85.0	89.8	+0.2 [−2.6, 3.0]	+0.2 [−2.8, 3.2]	−4.8 [−7.8, −2.0]
MMLU-Pro	43.1	41.9	50.0	49.0	+6.9 [3.1, 11.0]	+8.1 [4.0, 12.1]	+1.0 [−2.6, 4.5]
ARC-C	62.5	63.8	66.1	67.5	+3.6 [2.0, 5.3]	+2.3 [0.9, 3.7]	−1.4 [−2.6, −0.1]
BBH	66.6	71.3	73.5	71.7	+6.9 [4.6, 9.2]	+2.2 [0.1, 4.3]	+1.9 [−0.4, 4.1]
HumanEval	60.4	55.5	61.0	59.1	+0.6 [−6.7, 7.9]	+5.5 [−0.6, 11.6]	+1.8 [−5.5, 8.5]
MBPP+	74.3	79.4	79.1	79.4	+4.8 [1.3, 8.2]	−0.3 [−3.7, 2.9]	−0.3 [−3.7, 2.9]
HELMET	44.0	49.5	57.0	65.7	+13.0 [6.7, 19.5]	+7.5 [1.4, 13.8]	−8.7 [−13.3, −4.1]
Ouro-1.4B-Thinking							
GSM8K	86.4	89.5	89.5	92.3	+3.0 [1.3, 4.8]	+0.0 [−1.5, 1.4]	−2.9 [−4.3, −1.5]
MATH-500	76.4	77.6	75.6	85.2	−0.8 [−4.6, 3.0]	−2.0 [−5.8, 1.8]	−9.6 [−12.8, −6.2]
MMLU-Pro	39.8	34.8	43.3	46.2	+3.6 [−1.0, 8.1]	+8.6 [4.3, 12.9]	−2.9 [−6.7, 1.2]
ARC-C	56.1	57.7	60.4	62.5	+4.3 [2.1, 6.4]	+2.7 [1.0, 4.5]	−2.0 [−3.7, −0.4]
BBH	48.7	24.0	32.8	70.0	−15.9 [−18.9, −13.0]	+8.8 [6.2, 11.5]	−37.2 [−39.9, −34.5]
HumanEval	34.1	33.5	34.1	32.9	+0.0 [−6.1, 6.1]	+0.6 [−5.5, 6.7]	+1.2 [−4.9, 7.3]
MBPP+	70.4	74.3	72.2	75.1	+1.9 [−2.1, 5.8]	−2.1 [−5.8, 1.9]	−2.9 [−6.3, 0.5]
HELMET	45.0	39.2	46.5	64.9	+1.5 [−4.8, 7.8]	+7.3 [2.2, 12.7]	−18.3 [−22.8, −13.6]

Table 4: **Figure 4 in numbers, and on Ouro-1.4B-Thinking.** Per checkpoint, one loop adapter serves prefill 2 / decode 2 and prefill 4 / decode 2; same benchmarks, protocols and items for both checkpoints. Differences are score points on a 0–100 scale; brackets are paired 95% bootstrap intervals.

task / model	config	acc. (%)	time (s)
GSM8K Ouro-2.6B-T	native 2/2	95.3	70.0
	native 4/4	93.8	114.2
	prefill 2 / decode 2	96.9	62.6
	prefill 4 / decode 2	96.9	63.1
MATH-500 Ouro-2.6B-T	native 2/2	85.0	243
	native 4/4	82.5	469
	prefill 4 / decode 2	85.0	298
GSM8K SenseNova	native 2/2	98.4	48.0
	prefill 1 / decode 1	96.9	28.4
	prefill 2 / decode 1	95.3	30.6

Table 5: **End to end per answer**, prefill included (batch 1, one B300; Ouro-2.6B-Thinking and SenseNova-Loop-9B; 64 GSM8K and 40 MATH-500 problems, budgets 4,096 and 8,192 tokens). On GSM8K, prefill 4 / decode 2 spends 1.27× the decode time per generated token of native 2/2 at batch 1, but its answers are shorter.

8K context, 32 steps	n=1	n=2	n=3	n=4
native n/n	834	342	200	125
prefill 4 / decode d, one prompt cache	813	406	271	203
1K context, 2,048 steps		n=2		n=4
native n/n		586		219
prefill 4 / decode d, one prompt cache		570		253

Table 6: **Decode throughput** of Ouro-2.6B (tokens/s, best over tested batch sizes, one B300): native model against the loop adapter prefilling with 4 loops with one stored prompt KV cache.

+5.5 [3.7, 7.4] on Lev-2.6B and +4.0 [2.3, 5.7] on Lev-3B.

H Prefill Loops in a Hybrid Model

SenseNova-Loop-9B stacks three Gated-DeltaNet layers per full-attention layer. Its prompt state therefore has two parts: the attention KV, which the decoder attends to, and each Gated-DeltaNet layer's convolution and recurrent state, a fixed-size summary of the prompt from which the decoder continues and which it keeps updating with its own tokens (the prefill stops before the last prompt token, which the decoder processes first). We give the one-visit decoder the second visit's state for one kind of layer and the first visit's for the other (Table 8). The loop adapter was trained only with both states from the first or both from the second visit, so we also retrain it, with the same recipe and seed, giving the decoder the second visit's state only in the Gated-DeltaNet layers for prefill-2 samples (below). These are experiments on one checkpoint; they show how the decoder responds to each state, not where information is stored.

Without training, either part alone recovers most of the gain of the full second-visit state (RULER-16K +50.2 and +50.5 against +66.6; MMLU-Pro +23.7 and +21.5 against +36.2). With the trained loop adapter the two parts behave differently. The second-visit Gated-DeltaNet state raises long-context

variant	p/d	task	difference
maps frozen at identity†	4/1	RULER-16K	-11.4 [-14.7, -8.2]
	4/1	GSM8K	-1.7 [-3.6, 0.4]
	4/1	MMLU	-0.2 [-0.6, 0.1]
	4/2	RULER-16K	-2.6 [-4.5, -0.7]
	4/2	GSM8K	-0.2 [-1.5, 1.1]
	2/2	RULER-16K	-3.4 [-5.6, -1.3]
unrestricted keys‡	–	RULER-16K	all CIs include 0

Table 7: **Design variants** on Ouro-2.6B: variant minus our loop adapter at prefill m / decode n (full sets, paired). Differences are score points on a 0–100 scale; brackets are paired 95% bootstrap intervals. †Same data, budget, seed and (m, n) sampling; only the maps are not trained (LoRA only, the decoder reads the last prefill loop's KV directly). ‡Unrestricted key maps with one prompt KV cache per decode loop, at prefill 1 / decode 1, 2/2, 3/1, 3/2 and 4/2.

scores: HELMET $+9.1$ [6.5, 11.6] and RULER-16K $+4.9$ [2.1, 7.7], against -0.2 and $+0.7$ for the second-visit attention KV, which lowers RULER-4K (-4.5 [-7.2, -2.0]). The second-visit attention KV raises MMLU-Pro ($+4.6$ [2.0, 7.2]; Gated-DeltaNet state alone $+2.0$ [-0.4, 4.5]). The ARC-Challenge loss of the full second-visit state (Section 4.1) appears with the Gated-DeltaNet part (-2.7 [-4.5, -0.9]; attention KV alone $+0.8$). In the "GDN state" condition the decoder still attends to the first visit's KV, consistent with exact retrieval relying on attention in hybrid models (Afendulev et al., 2026); what improves is the recurrent state it continues from. On Ouro, which has only attention layers, KV from more prefill loops likewise helps knowledge and loses exact recall (Appendix J). The same trained loop adapter, given the second-visit state only in its Gated-DeltaNet layers, scores above the full second-visit state on HELMET by 5.6 points and on RULER-4K/8K/16K by 6.5, 4.2 and 3.7, shows no detectable difference on ARC-Challenge, and scores below it on MMLU-Pro by 3.5. The loop adapter retrained on this mixed state confirms the pattern within its own training distribution: against its own prefill 1 / decode 1 it gains on HELMET ($+7.2$ [4.7, 9.7]) and RULER-16K ($+6.1$ [3.2, 9.1]) and loses ARC-Challenge (-2.8 [-4.7, -0.9]); against the original loop adapter at prefill 2 / decode 1 it is better on HELMET ($+4.4$ [2.7, 6.2]) and RULER-4K/8K/16K ($+6.7$, $+2.8$, $+2.9$) and worse on MMLU-Pro (-2.7 [-5.0, -0.3]). Which state to give the decoder is therefore a task-dependent trade-off that holds with and without training for it.

Table 9 is the SenseNova-Loop-9B counterpart of Figure 4, at one decode visit through the repeated block instead of two. With the same loop adapter, prefilling with two visits instead of one raises MMLU-Pro ($+5.5$), BBH ($+7.0$) and HELMET ($+3.5$), lowers ARC-Challenge (-2.9), and leaves GSM8K, MATH-500 and HumanEval within noise. Against native 2/2, prefill 2 / decode 1 shows no detectable difference on GSM8K (-0.6 [-1.6, 0.4]; the $0.59\times$ answer time of Section 4.3 is measured for prefill 1 / decode 1) and scores lower on the other benchmarks, most on HELMET (-16.8); giving the decoder the second-visit state only in the Gated-DeltaNet layers narrows that HELMET gap to -11.2 [-13.0, -9.4]. The native one-visit model scores far lower on every benchmark.

I SenseNova-Loop-9B Without Training

SenseNova-Loop-9B's one-loop decoder collapses on its own. Handing that unchanged decoder the prompt state of a two-visit prefill—attention KV and the recurrent state of its linear-attention layers, no parameter changed—lifts RULER-4K from 21.6 to 83.5 and MMLU-Pro from 18.9 to 55.1, whereas GSM8K accuracy only rises from 2.2 to 10.9, far below the 95.8 of native 2/2: a second prefill visit recovers knowledge and long-context accuracy, not multi-step reasoning.

J Where More Prefill Loops Do Not Help

HELMET shows where more prefill loops help and where they hurt (Table 10). On both Ouro reasoning models, prefilling with four loops instead of one improves retrieval-augmented QA by 15 points and re-ranking by 13–18, but lowers exact recall from a JSON key–value store on Ouro-1.4B-Thinking (-23.5); against native 4/4, prefill 4 / decode 2 gains on retrieval-augmented QA and in-context learning ($+6.1$, $+6.6$) and loses exact recall (58.8 vs. 100), so its HELMET average is lower (-8.7 [-13.3, -4.1]). SenseNova-Loop-9B does not follow the Ouro pattern on recall and re-ranking. The recall category is small (34 items on Ouro).

More prefill loops do not help everywhere (Table 11). At one decode loop it gives a negative but inconclusive estimate on RULER-4K, where answers are copied verbatim, and clearly lowers

	untrained one-visit decoder			trained loop adapter		
benchmark	attention KV	GDN state	both	attention KV	GDN state	both
RULER-4K	+41.3 [37.2, 45.5]	+48.0 [44.0, 52.0]	+61.9 [57.8, 65.8]	−4.5 [−7.2, −2.0]	+2.7 [−0.5, 6.0]	−3.8 [−7.6, 0.2]
RULER-8K	+49.0 [44.8, 52.9]	+49.6 [45.7, 53.7]	+64.2 [60.0, 68.3]	−2.6 [−5.1, −0.2]	+2.3 [−0.8, 5.6]	−1.8 [−4.8, 1.3]
RULER-16K	+50.2 [46.0, 54.1]	+50.5 [46.4, 54.6]	+66.6 [62.7, 70.5]	+0.7 [−1.4, 2.8]	+4.9 [2.1, 7.7]	+1.2 [−1.7, 4.0]
MMLU-Pro	+23.7 [20.3, 27.0]	+21.5 [18.4, 24.8]	+36.2 [32.8, 39.8]	+4.6 [2.0, 7.2]	+2.0 [−0.4, 4.5]	+5.5 [2.9, 8.1]
ARC-C	+5.2 [3.2, 7.3]	+7.7 [4.9, 10.4]	+9.6 [6.7, 12.5]	+0.8 [−0.6, 2.1]	−2.7 [−4.5, −0.9]	−2.9 [−4.8, −1.0]
HELMET	+32.1 [30.1, 34.1]	+37.4 [35.5, 39.3]	+51.8 [49.8, 54.0]	−0.2 [−2.7, 2.2]	+9.1 [6.5, 11.6]	+3.5 [0.8, 6.1]

Table 8: **Which part of a hybrid model's prompt state benefits from a second prefill visit.** SenseNova-Loop-9B, one decode visit, after a two-visit prefill. Within the repeated block, "attention KV" gives the decoder the second visit's attention KV and the first visit's Gated-DeltaNet (GDN) convolution and recurrent states; "GDN state" the reverse; "both" the second visit's states throughout. Layers outside the block keep the two-visit prefill's states. Differences are to the one-visit prefill with the same decoder, in score points; brackets are paired 95% bootstrap intervals.

	score				prefill 2 / decode 1 minus	
benchmark	native 1/1	prefill 1 / decode 1	prefill 2 / decode 1	native 2/2	prefill 1 / decode 1	native 2/2
GSM8K	2.2	95.6	95.2	95.8	−0.4 [−1.4, 0.6]	−0.6 [−1.6, 0.4]
MATH-500	3.2	88.0	89.8	94.8	+1.8 [−1.0, 4.6]	−5.0 [−7.2, −2.8]
MMLU-Pro	18.9	60.3	65.8	68.2	+5.5 [2.9, 8.1]	−2.3 [−4.4, −0.2]
ARC-C	52.4	66.0	63.1	66.9	−2.9 [−4.8, −1.0]	−3.8 [−5.3, −2.4]
BBH	2.6	44.5	51.5	57.6	+7.0 [5.9, 8.0]	−6.1 [−7.0, −5.2]
HumanEval	3.0	75.0	76.2	81.7	+1.2 [−4.9, 7.3]	−5.5 [−11.0, 0.0]
HELMET	7.2	56.9	60.3	77.1	+3.5 [0.8, 6.1]	−16.8 [−18.7, −15.0]

Table 9: **SenseNova-Loop-9B at one decode visit.** One loop adapter serves prefill 1 / decode 1 and prefill 2 / decode 1; native 2/2 runs the repeated block twice per token (56 layer executions instead of 32). Same protocols as Table 1 (MMLU-Pro 980, HELMET 6,646, BBH 6,511 items). MBPP+ is omitted: its prompt forces the answer to start with code, and this thinking model then opens a reasoning block inside the code (native 1/1: 294, prefill 1 / decode 1: 236, prefill 2 / decode 1: 372, native 2/2: 375 of 378 answers), so its score measures this format failure. Differences are score points on a 0–100 scale; brackets are paired 95% bootstrap intervals.

comparison	RAG	re-rank	ICL	recall
2.6B-T, prefill 4 − 1	+15.4	+13.4	+5.1	−2.9
1.4B-T, prefill 4 − 1	+14.5	+18.0	+9.3	−23.5
SenseNova, prefill 2 − 1	+1.2	−32.9	+21.6	+24.0
2.6B-T, 4/2 − native 4/4	+6.1	−6.3	+6.6	−41.2

Table 10: **HELMET by category** (score differences on a 0–100 scale; prefill comparisons at one decode loop, same loop adapter). More prefill loops help retrieval-augmented QA (RAG), re-ranking and in-context learning (ICL) on Ouro but can lose exact key–value recall. Items: 700 / 41 / 333 / 34 per category on Ouro, 4,200 / 246 / 2,000 / 200 on SenseNova; intervals are in `paper_numbers.json`.

4-shot MATH-500 on Ouro-2.6B; without exemplars the same comparison turns positive on two model sizes and two decode loop counts. Removing exemplars also changes length and format, so we read this as a prompting interaction rather than as evidence that literal information is lost; linear probes of the cache, which find the next token more and (in Ouro) the previous token less linearly decodable after more loops, are consistent with it (Appendix K).

model	decode n	task	prefill 4 − prefill n
Ouro-2.6B	1	RULER-4K	−3.8 [−8.2, 0.3]
	1	MATH, 4-shot	−11.2 [−15.4, −6.8]
	1	MATH, 0-shot	+7.0 [3.0, 10.8]
	2	MATH, 0-shot	+7.0 [3.0, 11.2]
Ouro-1.4B	1	MATH, 4-shot	−4.0 [−8.2, 0.0]
	1	MATH, 0-shot	+4.8 [1.4, 8.4]

Table 11: **The effect of prefill loops depends on context length and prompting.** Same loop adapter, base models, prefill 4 / decode n minus prefill n / decode n; MATH-500 scored with math-verify on the full answer. Differences are score points on a 0–100 scale; brackets are paired 95% bootstrap intervals.

Other design variants. Sharing generated-token KV across decode loops ("shared answer KV", Appendix A) shows no significant difference at two and three decode loops and costs 2.6 GSM8K points at four. A gated branch over the first prefill loop's KV improves RULER-4K

by 2.3 points at prefill 4 / decode 1, leaves GSM8K unchanged within noise, and does not remove the 4-shot MATH deficit. (Ouro-2.6B, full sets: shared answer KV at 4/2, 3/3, 4/4: $+0.3$ [−1.5, 2.1], -0.8 [−2.4, 0.8], -2.6 [−4.2, −1.1]; first-loop branch at 4/1: RULER-4K $+2.3$ [1.0, 3.6], GSM8K $+0.7$ [−0.8, 2.2].)

K Additional Analyses

Unfinished answers of a one-loop decoder. A reasoning model written with one loop often gives no answer. On Ouro-2.6B-Thinking, native 1/1 gives no boxed answer on 43.1% of GSM8K problems; 43.5% of its answers approach the 4,096-token budget. Training the loop adapter at prefill 1 / decode 1 reduces answers without a boxed answer to 19.5%; with the same decoder, prefilling with four loops reduces them further to 12.7% overall, lower in every difficulty bucket (Figure 8, left), and shortens answers by 20% overall. A lexical hesitation-marker rate does not fall (12.2 vs. 13.8 per 1,000 tokens), so shorter answers do not come from fewer markers of hesitation; these counts do not identify the underlying reasoning process. The same decoder on SenseNova-Loop-9B, handed the two-visit prompt state without training, leaves no boxed answer on 43.7% of GSM8K problems instead of 96.8% (Figure 8, right).

What each loop stores. What changes in the cache from loop to loop? We fit linear probes to the frozen native model's per-loop value cache—what attention reads back from the prompt—at the same layer and with the same weights, varying only the loop (Figure 9). From loop 1 to loop 4 of Ouro-2.6B the previous token becomes less linearly decodable (57.7% → 45.5%, lower in 12 of 13 probed layers) and the next token more (26.0% → 36.1%, higher in all 13); Ouro-1.4B and Ouro-2.6B-Thinking agree. In SenseNova-Loop-9B the next-token gain replicates in all six attention layers (24.1% → 29.1%) while the previous token is about as decodable at both visits (56.8% vs. 56.5%). These are decodability results, not proof of information loss or causal use (Merin, 2026), and the probes read the native model, not the adapted decoder; they resemble the across-layer trend of Voita et al. (2019), and per-loop probes of looped transformers go back to Yang et al. (2023).

L Reproducibility

`python ouro/scripts/paper_numbers.py compute` recomputes the paired differences, intervals, means, timing ratios and Section 4.3 statistics of this paper from the accompanying per-item records (1 MB) in seconds on a laptop, and compares them with expected values hard-coded in the script, which mirror the numbers printed here. `REPRODUCE.md` maps the remaining results (throughput, probes) to their scripts and gives the training and evaluation commands that regenerate the per-item records.

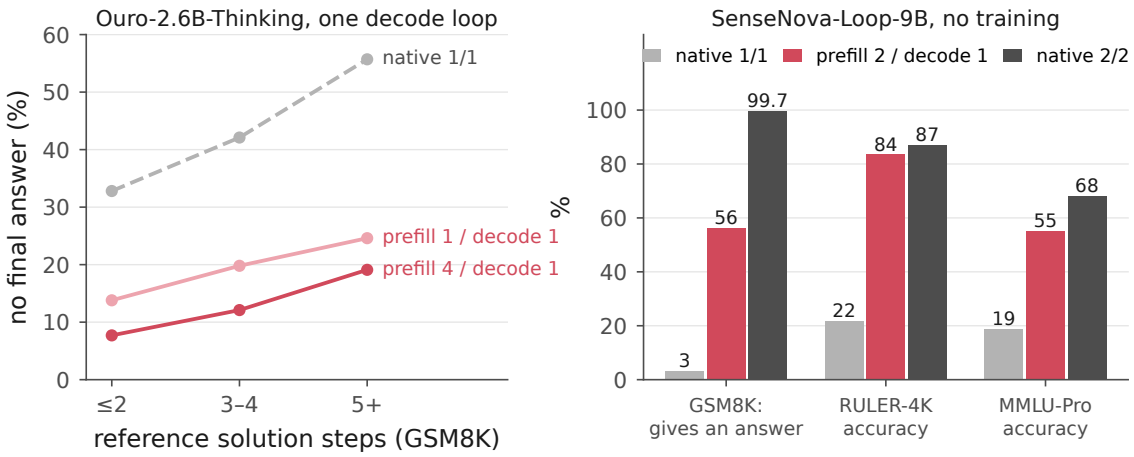


Figure 8: **More prefill loops help a one-loop decoder.** Left: Ouro-2.6B-Thinking at one decode loop, GSM8K problems bucketed by the number of lines in the reference solution (326 / 668 / 325 problems). Answers without a boxed answer grow with difficulty; prefilling with four loops instead of one lowers them by 5.5–7.7 points in every bucket. Right: SenseNova-Loop-9B without any training: the native one-loop decoder reads the prompt state of a two-visit prefill.

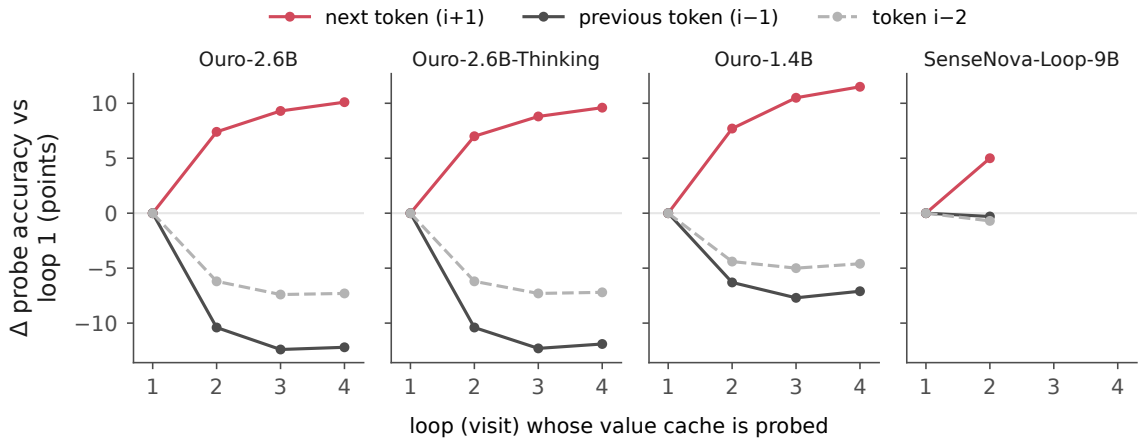


Figure 9: **Later loops make the next token more decodable; in Ouro the previous token becomes less decodable.** Linear probes on the stored value cache, same weights and layer, change from loop 1 (mean over probed layers; 384 FineWeb windows of 1,024 tokens, 20% held out; 2,048 most frequent training tokens).